

One-Day-Ahead 1% VaR Forecasting on QQQ Daily Log Returns

A stepwise model-selection study

Ethan Yao

Table of contents

1	Introduction	2
2	Historical Background	3
3	Purpose Statement and Mathematical Development	4
3.1	VaR and the forecasting objective	4
3.2	Rolling out-of-sample design	5
3.3	Violation indicator	5
4	Data Example with Representative Code	5
4.1	RiskMetrics variance recursion and VaR mapping	5
4.2	Gaussian or Student- t GARCH one-step VaR	6
4.3	GARCHX one-step forecast logic	6
4.4	GJR-GARCH variance recursion	6
4.5	Historical Simulation quantile extraction	6
4.6	CAViaR-SAV recursion and check-loss objective	7
4.7	FHS empirical residual quantile construction	7
4.8	Bootstrap one-step simulation	7
5	Stepwise Model Development and Comparison	8
5.1	5.1 Baseline Gaussian volatility models	8
5.1.1	Why start here?	8
5.1.2	RiskMetrics ($\lambda = 0.94$)	8
5.1.3	Gaussian GARCH(1,1)	8
5.1.4	What do the baseline results show?	9
5.2	5.2 Heavier-tailed parametric improvement: Student- t GARCH	9
5.2.1	Why is Student- t the next step?	9
5.2.2	Mathematical setup	10
5.2.3	Forecast intuition	10
5.2.4	Results	10
5.3	5.3 Extension path 1: GARCH-family extensions	11
5.3.1	First extension: GARCHX(VIX)	11

5.3.2	Second extension: GJR-GARCH	12
5.4	5.4 Extension path 2: Historical / direct-quantile alternatives	13
5.4.1	Historical Simulation	13
5.4.2	CAViaR-SAV	14
5.5	5.5 Final improvement: empirical-tail FHS / bootstrap methods	15
5.5.1	Stage 1: apply empirical tails to plain GARCH	16
5.5.2	Stage 2: apply empirical tails to GJR-GARCH	17
5.5.3	Final lesson from the stepwise path	18
6	Backtesting and Forecast Evaluation Metrics	18
6.1	Kupiec unconditional coverage test	18
6.2	Christoffersen independence test	19
6.3	Conditional coverage test	19
6.4	Check Loss	19
7	Forecast Uncertainty and 95% Confidence Intervals	19
7.1	Why point VaR forecasts need uncertainty measures	19
7.2	General quantile-CI logic	20
7.3	Model-specific CI constructions in this project	20
7.3.1	Gaussian parametric models	20
7.3.2	Student- t parametric models	20
7.3.3	FHS / empirical residual quantile models	20
7.3.4	Bootstrap models	21
7.3.5	Historical Simulation	21
7.3.6	CAViaR-SAV proxy interval construction	21
7.4	Why interval widths are informative but not perfectly identical across models	21
7.5	Interval-width comparison from the saved outputs	21
8	Integrated Final Comparison Table	22
9	Discussion	25
10	Limitations	26
11	Conclusion	26
12	References	27

1 Introduction

This report studies one-day-ahead 1% Value-at-Risk (VaR) forecasting for QQQ daily log returns. The goal is not to present a flat catalog of models. Instead, the report follows the way a modeler would actually work through the problem: begin with simple lecture-aligned baselines, check where they fail, use that evidence to motivate the next modeling choice, and continue until a convincing final model emerges.

That stepwise logic matters here because the central issue is not just volatility forecasting. For 1% left-tail risk, the main challenge is tail calibration. A model may forecast conditional variance

reasonably well and still produce poor VaR forecasts if its lower-tail distribution is too thin. The evidence in this project shows exactly that pattern. Gaussian volatility-based baselines are natural starting points, but they underestimate downside risk for QQQ. Student- t innovations help, and asymmetric volatility helps somewhat, but the biggest improvement comes from replacing parametric tail assumptions with empirical tail calibration through filtered historical simulation (FHS) or closely related bootstrap ideas.

Using the current saved project outputs, the final model-selection story is:

- start with **RiskMetrics** and **Gaussian GARCH(1,1)** as baseline volatility-VaR models,
- move to **Student- t GARCH(1,1)** when Gaussian tails prove too thin,
- explore two next directions: richer GARCH-family extensions and alternative nonparametric/direct-quantile approaches,
- then follow the strongest remaining path from Lecture 18 by using **empirical residual tails**,
- and arrive at **GJR-FHS** as the best final model, with **GARCH-FHS** very close behind.

The full integrated comparison uses the current saved forecast and backtest outputs with a common out-of-sample length of 848 trading days. When older notes conflict with the saved files, this report follows the saved outputs.

2 Historical Background

VaR became especially prominent in the 1990s as financial institutions looked for a compact summary of downside market risk. J.P. Morgan's *RiskMetrics* framework was a major practical milestone because it popularized daily variance-based VaR measurement using exponentially weighted volatility updating. That framework also strongly shaped classroom and industry treatments of market risk, which is why RiskMetrics remains a natural baseline in a project like this.

The econometric foundation behind modern VaR forecasting comes from conditional heteroskedasticity models. Engle's ARCH model and Bollerslev's GARCH extension formalized the idea that financial returns have time-varying conditional variance, with volatility clustering as a central stylized fact. Once volatility is modeled dynamically, one-step conditional quantiles can be constructed by combining the conditional mean, the conditional standard deviation, and an assumed or estimated innovation quantile.

As VaR use expanded, evaluation became just as important as construction. Kupiec (1995) proposed the unconditional coverage test, which checks whether the observed violation rate matches the nominal level. Christoffersen (1998) added the independence test, emphasizing that good VaR forecasts should not produce clustered exceptions. Their joint conditional coverage test remains a standard benchmark for VaR backtesting.

A second important development was the move away from purely Gaussian tails. Financial return tails are often heavier than Gaussian theory suggests, so it became common to use Student- t innovations or semi-parametric approaches. Barone-Adesi, Giannopoulos, and Vosper (1999) proposed Filtered Historical Simulation, which combines a volatility model with an empirical residual tail. This is especially relevant in the present project, because the final empirical lesson is that empirical tail calibration contributes more than simply changing the volatility recursion.

A different line of work models the conditional quantile directly instead of modeling the full conditional distribution first. Engle and Manganelli's CAViaR framework is the best-known example. It is theoretically appealing because VaR itself is the object of interest, but it also requires careful optimization and adequacy testing.

Finally, there is a broad literature on extending volatility models with external predictors. Here that role is played by VIX in a GARCHX benchmark. Economically, that is sensible: VIX reflects market-implied uncertainty. Empirically, though, the question is whether it improves one-day QQQ VaR enough to justify the extra structure.

3 Purpose Statement and Mathematical Development

3.1 VaR and the forecasting objective

Let r_t denote the QQQ daily log return,

$$r_t = \log \left(\frac{P_t}{P_{t-1}} \right).$$

For tail probability $\alpha = 0.01$, the one-day-ahead VaR is the conditional lower-tail quantile satisfying

$$\Pr(r_{t+1} < \text{VaR}_{t+1}(0.01) \mid \mathcal{F}_t) = 0.01.$$

So a 1% VaR forecast is a threshold that should be breached on roughly 1% of future days if the model is well calibrated.

For many models in this project, the return forecast has the location-scale form

$$r_{t+1} = \mu_{t+1} + \sigma_{t+1} z_{t+1},$$

which implies

$$\widehat{\text{VaR}}_{t+1}(\alpha) = \hat{\mu}_{t+1} + \hat{\sigma}_{t+1} \hat{q}_\alpha.$$

This decomposition is the organizing principle of the whole report. Different models differ in two ways:

1. how they forecast σ_{t+1} ,
2. how they obtain the tail quantile $\hat{q}_\alpha$.

The stepwise model-selection story is really the story of improving one or both of those components.

3.2 Rolling out-of-sample design

The saved integrated outputs use:

- 2,824 cleaned observations,
- a 70% initial training split,
- 848 one-step-ahead out-of-sample forecasts,
- expected 1% violations: 8.48.

At each forecast origin, the model is re-estimated using data available up to that date, then used to forecast next-day VaR. This rolling design is appropriate because VaR is a sequential risk-management tool, not an in-sample fitting exercise.

3.3 Violation indicator

Given a forecast $\widehat{\text{VaR}}_t$, define the violation indicator by

$$I_t = \mathbf{1}\{r_t < \widehat{\text{VaR}}_t\}.$$

If the model is correctly calibrated at the 1% level, the sequence I_t should have mean close to 0.01, and the hits should not cluster systematically.

4 Data Example with Representative Code

The project uses QQQ adjusted close data to form daily log returns, and VIX closing values only for the GARCHX benchmark. Since this is a modeling report rather than a workflow log, the representative code below focuses only on forecast construction and evaluation logic.

A useful way to frame the empirical study is that the same asset and same forecast horizon are examined under several increasingly refined VaR constructions. The code excerpts therefore illustrate the methodological differences directly.

4.1 RiskMetrics variance recursion and VaR mapping

```
sigma2 = np.var(r_train[:250])
for i in range(250, len(r_train)):
    sigma2 = lam * sigma2 + (1 - lam) * r_train[i] ** 2

sigma_next = np.sqrt(sigma2)
var_t1 = r_train.mean() + sigma_next * norm.ppf(alpha)
```

This is the classic EWMA volatility baseline: simple, historically important, and lecture-aligned.

4.2 Gaussian or Student- t GARCH one-step VaR

```
fit = arch_model(r_train * 100, p=1, q=1, dist="Studentst").fit(dispatch="off")
mu_hat = fit.params.get("mu", 0.0) / 100
sigma_hat = np.sqrt(fit.forecast(horizon=1, reindex=False).variance.values[-1, 0]) / 100
nu_hat = fit.params["nu"]
q_alpha = t.ppf(alpha, df=nu_hat) * np.sqrt((nu_hat - 2) / nu_hat)
var_t1 = mu_hat + sigma_hat * q_alpha
```

The Gaussian version is the same except that $q_\alpha = \Phi^{-1}(\alpha)$.

4.3 GARCHX one-step forecast logic

```
y_adj = r_train - r_train.mean()
fit_garchx = garchx(y_adj, order=c(1, 1), xreg=vix_train)
pred_var = predict(fit_garchx, n.ahead=1, newxreg=vix_t)
sigma_next = sqrt(pred_var)
var_t1 = r_train.mean() + sigma_next * qnorm(alpha)
```

The exogenous regressor is the current VIX input used to help forecast next-day conditional variance.

4.4 GJR-GARCH variance recursion

```
fit = arch_model(r_train * 100, p=1, o=1, q=1, dist="normal").fit(dispatch="off")
sigma_hat = np.sqrt(fit.forecast(horizon=1, reindex=False).variance.values[-1, 0]) / 100
z_hat = fit.std_resid.dropna().to_numpy()
q_emp = np.quantile(z_hat, alpha)
var_t1 = fit.params.get("mu", 0.0) / 100 + sigma_hat * q_emp
```

The important part is the asymmetric volatility structure from the $o = 1$ term. The last two lines show the FHS version, which becomes the winning path later in the report.

4.5 Historical Simulation quantile extraction

```
tail_sample = r_train[-250:]
var_t1 = np.quantile(tail_sample, alpha)
```

This is deliberately simple and provides a serious benchmark precisely because it does not rely on a volatility model.

4.6 CAViaR-SAV recursion and check-loss objective

```
def sav_path(beta, r, v0):
    v = np.empty(len(r))
    v[0] = v0
    for t in range(1, len(r)):
        v[t] = beta[0] + beta[1] * v[t-1] + beta[2] * abs(r[t-1])
    return -v

u = r - q_path
loss = np.sum(u * (alpha - (u < 0).astype(float)))
```

This shows the direct-quantile idea: the model estimates the quantile itself rather than volatility first.

4.7 FHS empirical residual quantile construction

```
fit = arch_model(r_train * 100, p=1, q=1, dist="normal").fit(dispatch="off")
mu_hat = fit.params.get("mu", 0.0) / 100
sigma_hat = np.sqrt(fit.forecast(horizon=1, reindex=False).variance.values[-1, 0]) / 100
z_hat = fit.std_resid.dropna().to_numpy()
q_emp = np.quantile(z_hat, alpha)
var_t1 = mu_hat + sigma_hat * q_emp
```

This is the core Lecture 18 idea used in the final best models.

4.8 Bootstrap one-step simulation

```
boot_z = np.random.choice(z_hat, size=B, replace=True)
boot_returns = mu_hat + sigma_hat * boot_z
var_t1 = np.quantile(boot_returns, alpha)
```

For one-step forecasting, bootstrap is closely related to FHS because both recover the tail from the empirical residual distribution.

5 Stepwise Model Development and Comparison

5.1 5.1 Baseline Gaussian volatility models

5.1.1 Why start here?

The natural starting point is the lecture-aligned volatility-to-VaR pipeline: first forecast tomorrow's conditional variance, then map that forecast into VaR with a Gaussian quantile. The two baseline models are **RiskMetrics** and **Gaussian GARCH(1,1)**.

These are sensible starting points for three reasons.

1. They are historically central in market-risk practice.
2. They are mathematically transparent.
3. They separate the core question cleanly: if volatility is modeled reasonably well but the tail is Gaussian, is that enough for 1% VaR on QQQ?

5.1.2 RiskMetrics ($\lambda = 0.94$)

RiskMetrics uses the EWMA recursion

$$\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) r_{t-1}^2, \quad \lambda = 0.94,$$

with one-step VaR forecast

$$\widehat{\text{VaR}}_{t+1}(\alpha) = \hat{\mu}_{t+1} + \hat{\sigma}_{t+1} \Phi^{-1}(\alpha).$$

Its purpose in this report is not to win the comparison. Its purpose is to anchor the study with the simplest mainstream volatility-VaR benchmark.

5.1.3 Gaussian GARCH(1,1)

Gaussian GARCH(1,1) allows more flexible variance dynamics than RiskMetrics:

$$r_t = \mu + \varepsilon_t,$$

$$\varepsilon_t = \sigma_t z_t, \quad z_t \sim N(0, 1),$$

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2.$$

The one-step VaR mapping is again

$$\widehat{\text{VaR}}_{t+1}(\alpha) = \hat{\mu}_{t+1} + \hat{\sigma}_{t+1} \Phi^{-1}(\alpha).$$

The intuition is straightforward: if volatility clustering is the dominant issue, then GARCH should outperform a simpler EWMA benchmark.

5.1.4 What do the baseline results show?

The saved outputs show that Gaussian-tail volatility models are not enough.

Model	Violations	Rate	Kupiec p	Christof- fersen p	CC p	Check Loss	Inter- preta- tion
Risk- Metrics ($\lambda =$ 0.94)	18	2.12%	0.0043	0.3925	0.0117	0.3781	Too many breaches; Gaus- sian tail too shallow
GARCH- Gaussian	22	2.59%	0.0001	0.5970	0.0005	0.3700	Worse still; volatil- ity dynam- ics alone do not fix the tail problem

The lesson from the first step is already clear. Even when conditional volatility is updated dynamically, the Gaussian lower tail is too thin for QQQ one-day 1% VaR. This motivates the next improvement: keep the GARCH volatility structure, but replace the Gaussian innovation assumption with a heavier-tailed parametric alternative.

5.2 Heavier-tailed parametric improvement: Student- t GARCH

5.2.1 Why is Student- t the next step?

Once Gaussian baselines underperform, the most natural next step is not to abandon volatility modeling, but to admit that the innovation distribution may have heavier tails. Student- t GARCH keeps the same GARCH variance recursion while replacing Gaussian innovations with a Student- t law.

5.2.2 Mathematical setup

The model is

$$r_t = \mu + \varepsilon_t,$$

$$\varepsilon_t = \sigma_t z_t, \quad z_t \sim t_\nu \text{ (standardized to unit variance),}$$

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2.$$

Then the one-step VaR becomes

$$\widehat{\text{VaR}}_{t+1}(\alpha) = \hat{\mu}_{t+1} + \hat{\sigma}_{t+1} \hat{q}_\alpha^{(t)},$$

where $\hat{q}_\alpha^{(t)}$ is the fitted standardized Student- t 1% quantile.

5.2.3 Forecast intuition

This model answers a focused question: if the problem with the baseline is primarily tail thickness, can a heavier-tailed parametric distribution repair the VaR forecast without changing the volatility recursion itself?

5.2.4 Results

Model	Violations	Rate	Kupiec p	Christof- fersen p	CC p	Check Loss	Inter- preta- tion
GARCH(1,1) [Student- t]	13	1.53%	0.1480	0.1890	0.1482	0.3503	Better than Gaus- sian base- lines, but still not close enough to the target

Student- t GARCH clearly improves on the Gaussian baselines. The violation rate drops from 2.59% under Gaussian GARCH to 1.53%, and check loss also improves. That is meaningful evidence that tail thickness matters.

Still, this is not the end of the modeling story. The model remains noticeably above the 1% target and does not yet deliver the strongest overall backtesting performance. So Student- t innovations help, but they do not fully solve the forecasting problem.

At this stage, two natural directions remain:

1. enrich the **volatility model family**,
2. try **simpler alternative tail approaches** outside that main volatility-extension path.

5.3 5.3 Extension path 1: GARCH-family extensions

5.3.1 First extension: GARCHX(VIX)

5.3.1.1 Purpose and intuition

A natural response after Student- t GARCH is to ask whether the volatility forecast itself can be improved by incorporating market information external to QQQ returns. VIX is an obvious candidate because it summarizes market-implied uncertainty and is often interpreted as a forward-looking stress indicator.

5.3.1.2 Mathematical setup

The package-based benchmark used in the saved outputs can be written as

$$y_t = r_t - \mu,$$

$$\sigma_t^2 = \omega + \alpha y_{t-1}^2 + \beta \sigma_{t-1}^2 + \delta x_t,$$

where x_t is standardized VIX. The one-step VaR forecast is then

$$\widehat{\text{VaR}}_{t+1}(\alpha) = \mu + \hat{\sigma}_{t+1|t} \Phi^{-1}(\alpha).$$

The benchmark is estimated by Gaussian quasi-maximum likelihood in the `garchx` package, but its final VaR mapping is still Gaussian.

5.3.1.3 Results

Model	Violations	Rate	Kupiec p	Christof- fersen p	CC p	Check Loss	Inter- preta- tion
GARCHX(VIX) [garchx pkg]	15	1.77%	0.0424	0.2621	0.0679	0.3549	Useful bench- mark, but not persua- sive as an im- prove- ment path

The evidence does not support pushing further in this direction for this project. Although adding VIX is economically sensible, the saved results are weaker than plain Student- t GARCH and much weaker than the best later FHS models. So GARCHX is informative as a comparison benchmark, but it is not the path that produces the final best model.

5.3.2 Second extension: GJR-GARCH

5.3.2.1 Purpose and intuition

Another natural response is to enrich the variance recursion itself by allowing asymmetric volatility effects. In equity markets, negative returns often increase future volatility more than positive returns of the same size. This is the leverage effect, and GJR-GARCH is designed to capture it.

5.3.2.2 Mathematical setup

The GJR-GARCH recursion is

$$r_t = \mu + \varepsilon_t,$$

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \gamma \varepsilon_{t-1}^2 I(\varepsilon_{t-1} < 0) + \beta \sigma_{t-1}^2.$$

If $\gamma > 0$, negative shocks raise future volatility more strongly than positive shocks.

The VaR mapping remains

$$\widehat{\text{VaR}}_{t+1}(\alpha) = \hat{\mu}_{t+1} + \hat{\sigma}_{t+1} \hat{q}_\alpha,$$

with either Gaussian, Student- t , or later empirical residual quantiles.

5.3.2.3 Results before empirical-tail calibration

Model	Violations	Rate	Kupiec p	Christof- fersen p	CC p	Check Loss	Inter- preta- tion
GJR- GARCH	15	1.77%	0.0424	0.4621	0.0972	0.3450	Asym- metry helps the volatil- ity story, but not enough by itself

GJR-GARCH does not yet dominate Student- t GARCH on violation frequency, but its check loss is slightly better than plain Student- t GARCH, which suggests that asymmetry is adding useful information. So compared with GARCHX, this is the more promising extension path. It still is not the final answer, but it gives a stronger volatility foundation for the next stage.

5.4 5.4 Extension path 2: Historical / direct-quantile alternatives

5.4.1 Historical Simulation

5.4.1.1 Purpose and intuition

Historical Simulation is included as a deliberately simple nonparametric benchmark. Instead of estimating a volatility model or assuming a parametric innovation distribution, it asks a practical question: if the next-day tail is approximated by the recent empirical distribution of returns, how competitive is that simple forecast?

5.4.1.2 Mathematical setup

Using a rolling window of size $W = 250$,

$$\widehat{\text{VaR}}_{t+1}(\alpha) = \text{Quantile}_{\alpha}(r_{t-W+1}, \dots, r_t).$$

5.4.1.3 Results

Model	Violations	Rate	Kupiec p	Christof- fersen p	CC p	Check Loss	Inter- preta- tion
Historical- Simulation	12	1.42%	0.2529	0.1569	0.1910	0.4092	Simple, compet- itive, and im- portant as a bench- mark

Historical Simulation is a serious benchmark in this project. It is not the best model, but it is clearly competitive. That matters because it keeps the model-comparison story honest: extra structure must earn its place.

5.4.2 CAViaR-SAV

5.4.2.1 Purpose and intuition

CAViaR follows a different idea entirely. Instead of modeling volatility first and then converting it into VaR, it models the conditional quantile directly. This makes it closely tied to the forecasting target itself.

5.4.2.2 Mathematical setup

The implemented Symmetric Absolute Value (SAV) recursion updates a positive latent risk level by

$$v_t = \beta_0 + \beta_1 v_{t-1} + \beta_2 |r_{t-1}|,$$

and maps back to the left-tail return quantile by

$$q_t = -v_t.$$

Parameters are estimated by minimizing the quantile check-loss criterion,

$$L(\beta) = \sum_t \rho_\alpha(r_t - q_t(\beta)),$$

where

$$\rho_\alpha(u) = u(\alpha - I(u < 0)).$$

5.4.2.3 What the current implementation includes, and what it does not

The current project uses a simplified SAV-only implementation with light optimization, limited starting values, and no Dynamic Quantile adequacy test. So it should be described fairly as an exploratory direct-quantile benchmark, not as a full implementation of the original paper.

5.4.2.4 Results

Model	Violations	Rate	Kupiec p	Christof- fersen p	CC p	Check Loss	Inter- preta- tion
CAViaR- SAV	6	0.71%	0.3664	0.0297	0.0625	0.3400	Cover- age is conser- vative, but cluster- ing and imple- menta- tion limits make the evi- dence uncon- vincing

This result is subtle. The model is **not** a catastrophic failure in the saved outputs. In fact, its unconditional coverage and check loss are numerically strong. But the independence test fails, which means the exception timing is not convincing enough, and the simplified implementation makes it hard to push this direction further with confidence in the current project. So the correct conclusion is that the direct-quantile idea is interesting, but the present simplified CAViaR path is not yet persuasive enough to become the main recommendation.

5.5 5.5 Final improvement: empirical-tail FHS / bootstrap methods

At this point the evidence suggests the following:

- Gaussian baselines are too thin-tailed.
- Student- t helps, but not enough.
- GARCHX(VIX) is not a strong direction here.
- GJR-type asymmetry is more promising than GARCHX.
- Historical Simulation is a strong benchmark.
- Simplified CAViaR is interesting but not convincing enough as the main path.

That naturally leads to the final improvement direction from Lecture 18: keep the volatility model, but estimate the lower tail empirically rather than through a fixed parametric quantile.

5.5.1 Stage 1: apply empirical tails to plain GARCH

5.5.1.1 Purpose and intuition

Filtered Historical Simulation replaces the parametric innovation quantile with the empirical lower-tail quantile of standardized residuals. The idea is simple but powerful: let the volatility model handle scale dynamics, and let the residual data speak for the tail.

5.5.1.2 Mathematical setup

After fitting a volatility model, compute standardized residuals

$$\hat{z}_t = \frac{r_t - \hat{\mu}_t}{\hat{\sigma}_t},$$

then set

$$\hat{q}_{\alpha, \text{emp}} = \text{empirical } \alpha\text{-quantile of } \{\hat{z}_t\}.$$

The one-step VaR is

$$\widehat{\text{VaR}}_{t+1}(\alpha) = \hat{\mu}_{t+1} + \hat{\sigma}_{t+1} \hat{q}_{\alpha, \text{emp}}.$$

5.5.1.3 Results for plain GARCH empirical-tail variants

Model	Violations	Rate	Kupiec p	Christof- fersen p	CC p	Check Loss	Inter- preta- tion
GARCH- FHS	7	0.83%	0.5984	0.7327	0.8212	0.3499	Large im- prove- ment over Gaus- sian and Student- <i>t</i> para- metric GARCH

Model	Violations	Rate	Kupiec p	Christof- fersen p	CC p	Check Loss	Inter- preta- tion
GARCH- t-FHS	6	0.71%	0.3664	0.7698	0.6371	0.3611	Also strong; some- what more conser- vative
GARCH- Bootstrap	5	0.59%	0.1934	0.8075	0.4166	0.3523	Good but conser- vative
GARCH- t- Bootstrap	6	0.71%	0.3664	0.7698	0.6371	0.3648	Similar to Student- <i>t</i> FHS

This is the first major jump in performance. Relative to Gaussian GARCH, empirical-tail GARCH variants move much closer to the desired 1% hit rate and pass the main backtests far more comfortably. That strongly suggests the biggest missing ingredient was tail calibration.

5.5.2 Stage 2: apply empirical tails to GJR-GARCH

5.5.2.1 Purpose and intuition

If asymmetry added something useful earlier, then the next question is whether asymmetry and empirical tails work even better together.

5.5.2.2 Mathematical setup

The variance recursion is still GJR-GARCH,

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \gamma \varepsilon_{t-1}^2 I(\varepsilon_{t-1} < 0) + \beta \sigma_{t-1}^2,$$

but the tail quantile is now empirical rather than Gaussian or Student-*t*:

$$\widehat{\text{VaR}}_{t+1}(\alpha) = \hat{\mu}_{t+1} + \hat{\sigma}_{t+1} \hat{q}_{\alpha, \text{emp}}.$$

5.5.2.3 Results for GJR empirical-tail variants

Model	Violations	Rate	Kupiec p	Christof- fersen p	CC p	Check Loss	Inter- preta- tion
GJR- FHS	8	0.94%	0.8672	0.6961	0.9137	0.3402	Best overall final model
GJR-t- FHS	6	0.71%	0.3664	0.7698	0.6371	0.3537	Strong but more conser- vative

This final step completes the modeling journey. Compared with plain GARCH-FHS, adding asymmetry through GJR produces the cleanest final calibration: the violation rate is closest to the nominal 1%, and the Kupiec, independence, and conditional coverage tests all look excellent.

5.5.3 Final lesson from the stepwise path

The biggest overall gain comes from **empirical-tail calibration**, not just from making the volatility recursion more elaborate. Asymmetry helps somewhat, and Student- t helps somewhat, but the decisive improvement arrives when the tail is estimated empirically. That is why the report ends with **GJR-FHS** as the best final model.

6 Backtesting and Forecast Evaluation Metrics

The model-comparison story above relies on four evaluation tools. Each should be defined before being interpreted.

6.1 Kupiec unconditional coverage test

The null hypothesis is that the violation probability equals the target level α . If x is the number of violations in n forecasts and $\hat{\pi} = x/n$, the likelihood-ratio statistic is

$$LR_{uc} = -2 \log \left[\frac{(1 - \alpha)^{n-x} \alpha^x}{(1 - \hat{\pi})^{n-x} \hat{\pi}^x} \right].$$

A large p-value means the observed violation rate is plausible under the target 1% rate. A small p-value means the model is systematically too liberal or too conservative.

6.2 Christoffersen independence test

Correct hit frequency is not enough if violations arrive in clusters. The independence test treats the hit sequence as a first-order Markov chain and compares:

- a model where hits are independent through time,
- a model where hit probability depends on whether the previous day was a hit.

A low p-value indicates violation clustering and therefore poor dynamic calibration.

6.3 Conditional coverage test

The conditional coverage test combines the previous two ideas:

$$LR_{cc} = LR_{uc} + LR_{ind}.$$

It is the most natural summary test because a good VaR model should get both the overall hit rate and the timing pattern roughly right.

6.4 Check Loss

Because backtests based only on violations ignore the size of forecast errors, the project also uses quantile check loss. The actual formula used in the code is

$$\rho_\alpha(u) = u(\alpha - I(u < 0)), \quad u_t = r_t - \widehat{\text{VaR}}_t.$$

Equivalently,

$$\text{CheckLoss}_t = (r_t - \widehat{\text{VaR}}_t) (\alpha - I(r_t < \widehat{\text{VaR}}_t)).$$

For non-violation days, this becomes approximately $\alpha(r_t - \widehat{\text{VaR}}_t)$, so overly conservative forecasts still incur some loss. For violation days, the loss becomes approximately $(1 - \alpha)(\widehat{\text{VaR}}_t - r_t)$, so large missed tail events are penalized much more heavily. Lower check loss is therefore better.

This metric is useful because it evaluates the quality of the full quantile forecast, not just whether the realized return barely crossed the threshold.

7 Forecast Uncertainty and 95% Confidence Intervals

7.1 Why point VaR forecasts need uncertainty measures

A VaR forecast is an estimate, not a known population quantity. It depends on estimated volatility, estimated or empirical tail quantiles, and finite-sample information. Reporting only a point VaR hides the fact that the forecast itself is uncertain.

7.2 General quantile-CI logic

Across the project, the common starting point is the large-sample quantile standard error

$$\text{SE}(\hat{q}_\alpha) = \sqrt{\frac{\alpha(1-\alpha)}{nf(\hat{q}_\alpha)^2}},$$

which leads to the generic 95% interval

$$\hat{q}_\alpha \pm 1.96 \text{ SE}(\hat{q}_\alpha).$$

For location-scale VaR models,

$$\widehat{\text{VaR}}_{t+1}(\alpha) = \hat{\mu}_{t+1} + \hat{\sigma}_{t+1} \hat{q}_\alpha,$$

so the quantile interval maps directly into a VaR interval.

7.3 Model-specific CI constructions in this project

The saved `lower_bound` and `upper_bound` columns are not generated by one universal formula. They are model-specific.

7.3.1 Gaussian parametric models

For Gaussian models such as Gaussian GARCH, RiskMetrics, and the package-based GARCHX benchmark, the project uses the asymptotic standard error of the Gaussian quantile and maps it through

$$\hat{\mu} + \hat{\sigma} (\Phi^{-1}(\alpha) \pm 1.96 \text{ SE}(\hat{q}_\alpha)).$$

7.3.2 Student- t parametric models

For Student- t parametric models, the same logic is used but with the standardized Student- t density and quantile replacing the Gaussian ones.

7.3.3 FHS / empirical residual quantile models

For FHS models, the project takes the empirical α -quantile of standardized residuals, estimates quantile uncertainty from the residual sample, and then maps that interval through the one-step volatility forecast.

7.3.4 Bootstrap models

For bootstrap models, the project resamples standardized residuals, simulates one-step returns, takes the empirical 1% quantile of those simulated returns as VaR, and builds the interval directly from that simulated predictive sample.

7.3.5 Historical Simulation

Historical Simulation uses the same sample-quantile interval logic, but now the sample is the rolling window of recent raw returns rather than standardized residuals.

7.3.6 CAViaR-SAV proxy interval construction

CAViaR does not produce a parametric likelihood-based VaR interval in the current workflow. Instead, the project recenters an empirical return sample so that its 1% quantile matches the CAViaR point forecast, then applies sample-quantile interval logic to that shifted sample. This should be interpreted as a proxy interval rather than an exact inferential object of the same type used for parametric GARCH models.

7.4 Why interval widths are informative but not perfectly identical across models

Because the intervals are constructed differently across model classes, their widths are informative but not perfectly homogeneous objects. A narrower interval does not automatically mean a better model, and widths should be interpreted together with backtesting performance.

7.5 Interval-width comparison from the saved outputs

Using the saved `lower_bound` and `upper_bound` columns, the mean interval widths are:

Model	Mean 95% CI width
CAViaR-SAV	0.02064
Historical-Simulation	0.02064
GARCH-t-FHS	0.00783
GARCH-FHS	0.00688
GARCH(1,1)	0.00677
GJR-GARCH	0.00637
GJR-t-FHS	0.00586
GJR-FHS	0.00436
GARCH-Gaussian	0.00369
GARCH-t-Bootstrap	0.00368
RiskMetrics ($\lambda = 0.94$)	0.00368
GARCHX(VIX) [garchx pkg]	0.00353
GARCH-Bootstrap	0.00303

Three points are worth emphasizing.

1. **Historical Simulation and CAViaR-SAV have the widest intervals.** That reflects the noisiness of empirical extreme quantiles and, for CAViaR, the proxy nature of the interval construction.
2. **The leading models combine moderate width with good calibration.** GJR-FHS and GARCH-FHS do not simply minimize width; they balance stable interval behavior with strong backtesting.
3. **Narrow intervals are not automatically desirable.** Gaussian models have relatively narrow intervals, but their VaR points are too shallow, so those narrow intervals are not a sign of superior risk forecasting.

8 Integrated Final Comparison Table

Table 1 brings the full evidence together. It includes violation counts, backtest p-values, check loss, and a short interpretation for each model.

Model	Violations	Rate	Kupiec	Christof- fersen	CC	Total Check Loss	Concise interpre- tation
GJR- FHS	8	0.94%	Pass	Pass	Pass	0.3402	Best overall final model; strongest balance of cali- bration and stability
GARCH- FHS	7	0.83%	Pass	Pass	Pass	0.3499	Very close second; empiri- cal tails matter more than extra com- plexity

Model	Violations	Rate	Kupiec	Christof- fersen	CC	Total Check Loss	Concise interpre- tation
GARCH- t-FHS	6	0.71%	Pass	Pass	Pass	0.3611	Strong but some- what conser- vative
GJR-t- FHS	6	0.71%	Pass	Pass	Pass	0.3537	Strong empirical- tail asym- metric model, also conser- vative
GARCH- t- Bootstrap	6	0.71%	Pass	Pass	Pass	0.3648	Similar to Student- <i>t</i> FHS; good but not the cleanest final choice
GARCH- Bootstrap	5	0.59%	Pass	Pass	Pass	0.3523	Conser- vative; useful support for empirical- tail logic
Historical- Simulation	12	1.42%	Pass	Borderline	Pass	0.4092	Simple and compet- itive; impor- tant bench- mark

Model	Violations	Rate	Kupiec	Christof- fersen	CC	Total Check Loss	Concise interpre- tation
GARCH(1,1) [Student- t]	13	1.53%	Pass	Pass	Pass	0.3503	Clear im- prove- ment over Gaus- sian tails, but not enough
CAViaR- SAV	6	0.71%	Pass	Fail	Borderline	0.3400	Interest- ing direct- quantile result, but not convinc- ing enough yet
GJR- GARCH	15	1.77%	Fail	Pass	Borderline	0.3450	Asym- metry helps some- what, but tail still not cali- brated well enough
GARCHX(VIX) [garchx pkg]	15	1.77%	Fail	Pass	Borderline	0.3549	Honest compar- ison bench- mark, not a recom- mended final model

Model	Violations	Rate	Kupiec	Christof- fersen	CC	Total Check Loss	Concise interpre- tation
Risk- Metrics ($\lambda =$ 0.94)	18	2.12%	Fail	Pass	Fail	0.3781	Histori- cally impor- tant baseline, but too thin- tailed here
GARCH- Gaussian	22	2.59%	Fail	Pass	Fail	0.3700	Clearly too thin- tailed for QQQ 1% left-tail risk

The table reinforces the main narrative. The final ranking is not driven by one metric alone. Instead, it reflects the cumulative evidence that empirical-tail GARCH models dominate simple Gaussian volatility-VaR constructions.

9 Discussion

The model-comparison exercise points to a fairly clean methodological lesson. For one-day-ahead 1% VaR on QQQ, the hardest part is not simply forecasting volatility; it is forecasting the lower tail well enough once volatility is forecasted.

The first stage of the study shows that classic Gaussian volatility-to-VaR methods are too weak. RiskMetrics and Gaussian GARCH are reasonable baselines, but they systematically underestimate downside risk. The second stage shows that Student- t innovations improve matters in the expected direction. That is evidence that heavier tails are essential. The third stage shows that richer volatility structures alone are not the main answer. GARCHX(VIX) is not persuasive in the saved results, and GJR-GARCH only helps somewhat. The fourth stage shows that Historical Simulation remains a serious benchmark and that direct-quantile modeling through simplified CAViaR is interesting but not yet convincing enough. The final stage shows that once the tail is estimated empirically, performance improves sharply.

So the main substantive lesson is simple: **the biggest gain comes from empirical tail calibration, not just from changing the variance recursion.**

A second useful lesson is that asymmetry still matters, just less than tail treatment. GJR-FHS beats GARCH-FHS narrowly but consistently, which suggests that asymmetry is worth incorporating

once the tail is handled properly. That is exactly the kind of incremental modeling logic the report set out to show.

10 Limitations

Several limitations should be stated directly.

1. **The final comparison is integrated from multiple saved outputs.** The analysis is internally consistent because all included models share the same 848-day out-of-sample horizon, but the repository workflow is not a single unified runner.
2. **GARCHX is a benchmark, not a polished final candidate.** The package-based implementation is cleaner than the earlier custom version, but the final VaR mapping is still Gaussian, so it should not be overinterpreted.
3. **The CAViaR implementation is simplified.** Only the SAV form is used, optimization is relatively light, and the Dynamic Quantile adequacy test from the original CAViaR framework is not included.
4. **Bootstrap forecasts are mechanically deterministic in the saved implementation.** The code resets the random seed inside the forecast function, which improves reproducibility but makes the rolling bootstrap less genuinely stochastic than a fresh-resampling implementation would be.
5. **The conclusions are application-specific.** The ranking is for QQQ, one-day horizon, and 1% VaR. Other assets, horizons, or tail levels could change the ordering.

11 Conclusion

This report reorganized the VaR study as a stepwise model-selection story rather than a flat list of models. The starting baseline was a pair of Gaussian volatility-based VaR methods: **RiskMetrics** and **Gaussian GARCH(1,1)**. They were natural first choices because they are historically central, mathematically transparent, and closely aligned with standard lecture material. But the evidence showed that they were not enough. Both produced too many VaR breaches, which indicated that Gaussian left tails were too thin for QQQ one-day 1% risk forecasting.

The next improvement was **Student- t GARCH(1,1)**. This helped, confirming that heavier-tailed innovations matter, but it still did not fully solve the forecasting problem. After that, the report followed two directions. In the GARCH-family extension path, **GARCHX(VIX)** did not perform well enough to justify itself as the main direction, while **GJR-GARCH** suggested that asymmetry was somewhat useful. In the alternative-method path, **Historical Simulation** proved to be a strong benchmark, while the simplified **CAViaR-SAV** implementation was interesting but not convincing enough yet because of dependence issues and implementation limits.

The final improvement came from the Lecture 18 empirical-tail path. Applying **FHS and related bootstrap ideas** to plain GARCH already improved performance strongly. Applying the same idea to **GJR-GARCH** improved further and produced the best overall final model. The final recommendation is therefore **GJR-FHS**, with **GARCH-FHS** a close second.

The central methodological lesson is clear. For this application:

- Gaussian volatility-only VaR is too weak,
- Student- t helps,
- asymmetry helps somewhat,
- GARCHX is not persuasive enough here,
- Historical Simulation is a good benchmark,
- the simplified CAViaR path is not yet convincing,
- and the largest improvement comes from **empirical tail calibration**.

That is why the report ends with **GJR-FHS** as the best final model for one-day-ahead 1% VaR forecasting on QQQ daily log returns.

12 References

- Barone-Adesi, G., Giannopoulos, K., & Vosper, L. (1999). VaR without correlations for portfolios of derivative securities. *Journal of Futures Markets*, 19(5), 583–602.
- Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3), 307–327.
- Christoffersen, P. F. (1998). Evaluating interval forecasts. *International Economic Review*, 39(4), 841–862.
- Engle, R. F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, 50(4), 987–1007.
- Engle, R. F., & Manganelli, S. (2004). CAViaR: Conditional autoregressive value at risk by regression quantiles. *Journal of Business & Economic Statistics*, 22(4), 367–381.
- J.P. Morgan/Reuters. (1996). *RiskMetrics technical document* (4th ed.).
- Kupiec, P. H. (1995). Techniques for verifying the accuracy of risk measurement models. *Journal of Derivatives*, 3(2), 73–84.
- Pascual, L., Romo, J., & Ruiz, E. (2006). Bootstrap prediction for returns and volatilities in GARCH models. *Computational Statistics & Data Analysis*, 50(9), 2293–2312.